

Abstract

Most competition working notes describe a winning recipe. This note deliberately does the opposite, and argues the inversion is useful: a systematic, measurement-driven account of *why a strong shared public baseline could not be beaten with consumer-scale resources*, and what that reveals about the structure of modern bioacoustic competitions. By mid-competition a single architectural pattern — a frozen Perch v2 embedding model, a Light-ProtoSSM sequence head, a distilled Sound-Event-Detection (SED) CNN, and a taxonomy-smoothing post-processor — had been openly shared and converged on a public-leaderboard score of **0.950**, where (as of the public-LB snapshot on 2 June 2026) roughly **890 of 4243 teams scored in the [0.950, 0.951) band**. We treat this plateau as an object of study rather than a target. We built a *measurement-gate* framework that reduces each improvement hypothesis to a cheap offline quantity and tests it *before* spending one of five daily submission slots, and used it to test every lever available to a small team: own-trained SEDs, foundation-model linear probes, rare-taxon specialists, post-hoc calibration tricks, and architecturally-diverse model ensembles. Each was neutral or negative. Two observations we found practically consequential in this setting: (1) the public components’ apparent near-perfect held-out accuracy is *optimistic by construction* — they are fit on the only in-domain labelled audio available for validation — which inverts the most natural offline comparison and makes held-out macro-AUC an unreliable basis for ranking a challenger against the baseline; and (2) the binding constraint for a small team is not modeling skill but **platform limits** — a ~ 1 MB notebook-size cap and a 90-minute CPU runtime budget — which physically forbid the one technique with genuine headroom (ensembling multiple full foundation-model pipelines). On a single held-out split, architecturally-diverse CNN ensembling improved macro-AUC by **+0.021** over its best member (a leak-controlled, directional result, §5.6), yet that ensemble could not be deployed within the runtime envelope. We close with a hypothesis, offered explicitly as conjecture since the leaders’ solutions are private: in the foundation-model era a public pipeline assembled from large pretrained components plus a community-discovered post-processing trick is a remarkably hard floor, and we conjecture the residual gap to the leaders is paid largely in up-front training compute rather than inference-time cleverness. All code and the measurement-gate harness are released openly.

Keywords: *bioacoustics · BirdCLEF · foundation models · negative results · ensemble diversity · knowledge distillation · data leakage*

Mapping the Public Ceiling: Negative Results and a Platform-Constraint Hypothesis for BirdCLEF+ 2026

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1 Introduction

BirdCLEF+ 2026 asked participants to identify which of 234 species — birds, amphibians, mammals, reptiles, and insects — vocalize in 1-minute soundscapes from the Brazilian Pantanal, an important conservation-monitoring task. Submissions are CPU-only Kaggle notebooks evaluated on a hidden set of ~ 600 recordings, scored by macro-averaged ROC-AUC that skips classes with no true positives in the test set; the runtime budget is **90 minutes**, internet is disabled, and each team may submit five times per day.

The competition exhibited an unusually sharp *public-kernel monoculture*. A lineage of openly-shared notebooks (“Ensemble-of-Solutions”, EoS.x, and their forks) raised the public ceiling in discrete, community-wide jumps — $0.928 \rightarrow 0.948 \rightarrow 0.949 \rightarrow \mathbf{0.950}$ — each jump driven by one new mechanism that the field adopted within days. The cumulative public-LB distribution (snapshot 2 June 2026, 4243 teams) was top-heavy and then sharply flat: **11 teams** ≥ 0.96 , **42** ≥ 0.955 , **252** ≥ 0.951 , and **1140** ≥ 0.950 — i.e. ~ 890 teams whose best public score fell in the narrow $[0.950, 0.951)$ band, the large population running the shared pipeline essentially unchanged. (Public-LB counts drift as teams resubmit; we report one dated snapshot.)

We entered this monoculture late and spent the competition on one question: *can a small team, on a single consumer GPU plus a small cloud budget, add anything to the public 0.950 pipeline that survives contact with the hidden test?* Our answer is negative. Each hypothesis was reduced to a measured quantity and gated before a submission was spent. The contribution of this note is therefore not a model but a **map**: what we tried, the numbers, the structural reason each path failed, and a small reusable methodology for resource-constrained competitors. Negative results reported with measurements are comparatively rare in this venue, since most write-ups come from teams that succeeded; we hope the surrounding terrain is useful.

2 Related work

BirdCLEF has run annually since 2014; recent editions converged on transfer-learning from large audio/bird foundation models followed by sequence modeling and heavy ensembling of diverse CNNs (the consistent ingredient in winning solutions across 2023–2025) [5]. Our public baseline reflects this: Perch v2 [1] as the embedding backbone, a ProtoSSM-style selective-state-space [2] head over its embeddings, and a distilled SED CNN. Our own-model attempts draw on noisy-student self-training [7] (pseudo-labeling unlabeled soundscapes with a teacher, then training a student) and on foundation-model linear probing (Bird-MAE [6], AudioMAE [4]). Two themes we engage critically are (i) knowledge distillation’s ceiling [3] — a student trained on a teacher’s outputs is bounded by that teacher — and (ii) ensemble diversity — that decorrelated architectures, averaged, beat any single member. Our contribution relative to this literature is empirical and negative: we quantify *where each of these reliably-useful techniques stops working* under this competition’s specific data-leakage and platform constraints.

3 Anatomy of the public 0.950 pipeline

The converged public solution is a four-stage rank ensemble:

1. **Perch v2** [1] — Google’s bird-vocalization foundation model — produces a 1536-d embedding and 234-logit score per 5-second window. It is the single most expensive inference stage and the dominant signal.
2. **Light-ProtoSSM** — a small (≈ 0.75 –5.8 M-parameter) bidirectional selective-state-space [2] head with prototype classification and cross-window attention — is fit (or loaded pre-trained) on Perch embeddings of the labelled soundscapes, contributing $\sim 60\%$ of the rank blend.
3. **Distilled SED** — a 5-fold EfficientNet-B0 sound-event detector on 256×313 log-mel input, distilled [3] from a stronger teacher ensemble, contributing $\sim 40\%$.
4. **TAX_SMOOTHING** — the final-jump mechanism ($0.949 \rightarrow 0.950$): each species’ score is pulled slightly toward the mean of its genus- and class-siblings ($\alpha_{\text{genus}} \approx 0.15$, $\alpha_{\text{class}} \approx 0.05$), exploiting taxonomic correlation among co-occurring Pantanal taxa.

We verified directly that every public 0.950 notebook is a scalar-tuned variant of this stack: forks differing in post-processing (BirdNET sidecars, PCEN, out-of-fold gating, iterative smoothing) and even one with a different ProtoSSM implementation all returned exactly 0.950 or below.

4 Method: measurement gates before submission slots

The scarce resources here are not FLOPs but **submission slots** (five per day, each scored only after a multi-hour hidden-test rerun) and **trustworthy validation signal** (the labelled audio is tiny and, as §5.2 shows, optimistically biased toward the public models). Our central methodological move was to never spend a slot on an unmeasured hypothesis. For each idea we built an offline *gate* on cloud GPU (Modal) or the local GPU:

- **compare_to_teacher** — train a candidate SED with a clean *file-level* hold-out of 13 labelled soundscapes; run the public distilled SED on the *same* held-out files; report both macro-AUCs and the per-class win count.
- **eval_*_probe / eval_rare_taxa** — train linear probes on frozen foundation-model embeddings; evaluate against the public SED *sliced to specific taxa*; measure orthogonality (how many classes the candidate strictly beats the baseline on).
- **eval_sed_ensemble** — measure whether an ensemble’s macro-AUC exceeds its best single member (a leak-free decorrelation test, independent of any public teacher).

Each gate ran in minutes on a single cloud GPU at low cost (we did not log a per-gate cost breakdown; total cloud spend across the study was a few US dollars). Each falsified direction would otherwise have consumed one-to-many submission slots and a day of latency. We regard the leak-aware gate methodology (§9) as a small transferable contribution for code-competitions with daily quotas and slow scoring.

Table 1: Levers tested against the public 0.950 baseline. (“LB” = public leaderboard at submission time; held-out figures are single-split offline gates, §4.)

Lever	Measured outcome	Verdict	Mechanism of failure
Own SED, focal-trained	LB 0.931	−0.019	distilled copy of a weaker teacher
Own SED, soundscape noisy-student	LB 0.933	−0.017	bounded by the public SED’s own pseudo-labels
compare_to_teacher (clean 13-file hold-out)	student 0.910 vs public 0.994	—	but the 0.994 is leak-optimistic (§5.2)
Bird-MAE / AudioMAE linear probe	beats public on 0/27 classes	dominated	no orthogonal signal to blend
Rare-taxon (frog/insect) specialist	beats public on 0/16 rare classes	dominated	public SED already 0.991 on rare taxa
Post-hoc tricks (BirdNET, co-occurrence, genus-proxy, iterative smooth)	LB 0.948–0.950	neutral/neg.	perturb a tight calibrated equilibrium
Diverse-CNN SED ensemble (4 archs)	0.954 vs best single 0.933 (single split)	+0.021	gain real but undeployable (§6)
Diverse-ensemble <i>swapped</i> into pipeline	LB 0.945	−0.005	our members individually < public SED on hidden test
Diverse-ensemble <i>added</i> (7 or 9 SEDs)	runtime timeout	—	90-min CPU budget exceeded

5 Systematic negative results

Table 1 consolidates the experiments. Each row was decided by a gate (offline macro-AUC, leak-aware) and/or a leaderboard submission.

5.1 Own-trained SED cannot match the distilled public SED

We trained EfficientNet-B0 SEDs two ways: focal-recording-only (public LB **0.931**) and a noisy-student [7] variant on 10,592 pseudo-labelled soundscapes → 127,896 soft-labelled windows (public LB **0.933**). Both regressed ~ 0.017 – 0.019 below the baseline when blended in. The gate indicated the cause: on a single 13-file hold-out the public SED scored **0.994** versus our student’s **0.910** — though that 0.994 is optimistically biased (§5.2). A single own-trained CNN on one GPU does not reach a model distilled from a strong teacher ensemble — and, because our students learn from the public SED’s *own* pseudo-labels, they are its distilled copies, structurally bounded below it [3].

5.2 The validation trap: optimistic held-out for public components

The gate number above (public SED 0.994 on hold-out) is **misleading in a way worth making explicit**, because it determined how we validated everything afterward. The public Light-ProtoSSM is, by inspection of the public notebook, fit on the labelled soundscapes (it trains on them in-kernel); the distilled SED was plausibly distilled with them in scope. Either way, the public pipeline’s 0.994 on a held-out *subset of those same soundscapes* is optimistic by construction, whereas our genuinely-held-out student is scored fairly — an asymmetric comparison. The gap between the public components’ ~ 0.99 on labelled audio and the full pipeline’s 0.950 / 0.942 on the hidden test is consistent with this optimism. The practical consequence: **an offline comparison of a challenger against a public component on the labelled soundscapes is biased in the baseline’s favour**, so held-out macro-AUC cannot reliably rank a small team’s model against the shared pipeline. This is, of course, an instance of the textbook train/test-leakage pitfall; our contribution is not the concept but the demonstration that, in this competition, it silently invalidates the *most natural* validation a competitor would reach for, and the consequent discipline of preferring *leak-controlled relative* gates (e.g. ensemble-vs-best-single, §5.6) over absolute comparisons to the baseline. (Whether the precise mechanism is label leakage or in-domain over-fit plus distribution shift, the methodological conclusion is identical.)

5.3 Foundation-model probes are strictly dominated

Linear probes on Bird-MAE-Base [6] and AudioMAE [4] embeddings beat the public SED on **0/27** held-out classes with positives; every rank-blend weight monotonically *reduced* macro-AUC. A strictly-dominated

source cannot improve a calibrated ensemble — it can only dilute it.

5.4 Rare-taxon specialists: a plausible premise, falsified

We hypothesized the bird-tuned pipeline would be weak on rare *Amphibia/Insecta* — stationary frog/insect textures unlike bird syllables — and that a texture-oriented model (AudioMAE [4]) could win there. Sliced to 16 measurable rare classes (954 held-out positives), the gate refuted it: the public SED scored **0.991** on exactly these taxa (it had learned them from the labelled soundscapes), and our specialists beat it on **0/16** (Bird-MAE 0.876, AudioMAE 0.799 macro-AUC on the slice). External audio cannot rescue the data-starved tail either: of 53 classes with < 10 training recordings, Xeno-Canto (a bird archive) covered 1, and 28 — including the **sonXX** insect “morphotypes” — are competition-internal labels with no scientific name and thus no external data anywhere.

5.5 Post-hoc tricks perturb a tight equilibrium

Genus-proxy boosting, sonotype mirroring, per-taxon temperature, BirdNET 3-way blending, cross-class co-occurrence boosting, and iterative second-pass smoothing each layered onto the 0.950 base returned 0.948–0.950 (neutral-to-negative). The public pipeline’s calibration chain (priors, rank-aware scaling, power optimization, threshold tuning, taxonomy smoothing) is a tight equilibrium; post-hoc reorderings disturb it more than they help. Tellingly, the community’s own last +0.001 came not from *adding* a trick but from **TAX.SMOOTHING** applied *inside* the calibrated chain at the right scale.

5.6 Diverse-CNN ensembling: a real gain that cannot be deployed

The most reliable technique in the field is ensembling architecturally-diverse CNNs, and we used it as a *controlled internal* test rather than a comparison to the (leak-flattered) baseline. We trained four diverse SED backbones — EfficientNetV2-S, ConvNeXt-Small, EfficientNet-B3, SEResNeXt50. Crucially, the 13 hold-out files were excluded from **all** four members’ training, *including the soundscape pseudo-label generation* — so the ensemble-vs-member comparison is leak-controlled with respect to that split (both sides are equally held out, unlike the baseline of §5.2). On that single split the four-model rank ensemble scored **0.9542** versus the best single member’s **0.9332**, a **+0.021** improvement — the expected diversity effect, observed cleanly. We stress this is a *single-split point estimate*: we did not run multiple folds or seeds and report no confidence interval, so +0.021 should be read as directional corroboration of the well-known diversity benefit, not a precise effect size (see §7).

This still could not break 0.950, for two largely independent reasons that are the crux of this note. First, *deployment runtime*: adding the four SEDs to the public pipeline ($5 + 4 = 9$, or even $5 + 2 = 7$ inferences) exceeded the 90-minute budget and timed out — a wall independent of model quality. Second, *substitution cost*: swapping two public SEDs for two of ours (holding the count at 5, runtime-safe) scored **0.945** on the LB. We read this single point cautiously: it is *consistent with* our members being individually weaker than the public SED on the genuine hidden test (where the baseline’s leak advantage vanishes), but it is confounded — the rank-blend weights were not re-tuned for the substitution, and our members were only half-trained when a cloud-budget cap froze them, which independently predicts a drop. The runtime conclusion is therefore firm; the substitution conclusion is suggestive, not proven. What the two together establish is narrower but solid: *we could not realize the diversity gain within the platform envelope*, whether because of runtime (certain) or insufficient member quality (likely, given half-training) or both.

6 The platform-constraint finding

The deeper lesson generalizes beyond SEDs. The one strategy with real headroom over a converged baseline is to **ensemble multiple full, diverse pipelines**. We attempted to combine two genuinely different public 0.950 pipelines (distinct ProtoSSM implementations and auxiliary models) in one notebook. This is blocked by *platform physics*, not modeling:

- **Notebook-size cap (~ 1 MB).** The two full pipelines concatenated to 1.08 MB and were rejected at upload; a single pipeline is 0.77 MB.
- **Runtime cap (90 min CPU).** Each pipeline’s Perch pass nearly fills the budget; two passes time out — the same wall that killed our additive-SED ensembles.

These limits are, we argue, *a* structural reason for the monoculture: heavy foundation-model pipelines cannot be ensembled at inference inside the platform, so the public field — which mostly forks the same heavy pipeline — is funnelled into a single 0.950 cluster. We then *conjecture* (the leaders’ solutions are private, so this is not established) that the teams above 0.95 escape the funnel by pre-training their own lighter models that are individually competitive yet cheap enough to run several within 90 minutes, paying the cost up front in training compute that a late, small entrant on one consumer GPU cannot match. This conjecture has at least an empirical prior: published BirdCLEF 2023–2025 winning solutions [5] consistently rely on *ensembles of several independently-trained CNNs* run within the inference budget, rather than a single shared backbone. We present it as the most parsimonious hypothesis consistent with our measurements, not as a demonstrated fact about the 11 teams ≥ 0.96 .

7 Threats to validity

We state the limits of these conclusions plainly. (i) **Single-split, no error bars.** Our offline gates use one fold and a small hold-out (13 files; 32 classes with positives for the absolute gates, 16 rare classes for §5.4); we report point estimates without confidence intervals. The headline +0.021 ensemble gain (§5.6) is therefore directional corroboration of a known effect, not a measured effect size; multi-fold/multi-seed validation with CIs is the most important piece of future work, which our deleted cloud infrastructure prevented us from adding before submission. (ii) **The leak claim is partly inferential.** We observe that the public ProtoSSM trains on the labelled soundscapes in-kernel, but only infer the SED’s training scope from the 0.99-vs-0.95 optimism gap; the methodological conclusion holds under either label leakage or in-domain over-fit plus distribution shift. (iii) **Half-training cuts two ways.** The diverse ensemble was half-trained when a budget cap froze it. This understates its absolute ceiling (so the *deployment-wall* conclusion is conservative — a stronger ensemble would still time out), but it is also a genuine confound for the *substitution* result (§5.6): a fully-trained ensemble might have made the runtime-safe swap competitive, so we do not claim the swap result proves our members are intrinsically inferior. (iv) **The leaders are a conjecture.** Our characterization of the > 0.95 teams (§6) is unverified — their solutions are private. (v) **Scope.** All scores use this competition’s masked macro-AUC on its specific data; the *platform-constraint* and *validation-trap* lessons should transfer qualitatively to similar code-competitions, but the exact thresholds will not.

8 What moved, and final standing

The only positive movement available to us was tracking the community’s TAX_SMOOTHING jump to 0.950 and choosing robust final submissions. Recognizing the ~ 890 -team public tie, we selected two maximally-divergent 0.950 pipelines to hedge the public $\rightarrow$ private shift. The shift proved mild and uniform (0.950 $\rightarrow$ **0.942** across our picks); a small robustness signal favoured the more battle-tested calibration (the older EoS.8 base held 0.942 while the newer EoS.9 fell to 0.941). Final standing: **289 / 4243, bronze**.

9 Discussion

Three takeaways we believe are non-obvious and worth recording:

1. **Optimistic held-out can invert your conclusions.** When public components are fit on the only in-domain labelled data available for validation, offline comparisons systematically flatter them. Resource-constrained teams should treat such comparisons as *lower bounds* on their own models’ relative quality, and prefer leak-free relative gates.
2. **Leak-aware offline gating.** Validating locally before submitting is standard; the twist that mattered here is *what* to validate against. Because absolute held-out comparison to the baseline is biased (point 1), we gated on *relative* and *orthogonality* signals instead (ensemble-vs-best-single; per-class win counts against the public SED). This let us reach a decision on five directions within the budget a blind approach would spend on one, and we recommend the leak-aware variant specifically for code-competitions where the only in-domain labels were also used to fit the public baseline.
3. **In the foundation-model era the binding constraint is often the platform, not the idea.** The highest-value technique (pipeline ensembling) and a genuine +0.021 ensemble gain were both real and both undeployable under a 1 MB / 90-minute envelope. A public ceiling built from large shared pretrained components is not a wall of cleverness to out-think; it is a floor, and the door through it is pre-training compute a late, small team does not have.

Taken together these sharpen the question of where effort actually pays under foundation-model baselines and platform limits, and suggest that a useful contribution from a mid-pack team is a measured map of the terrain that winners’ recipes leave unspoken.

10 Conclusion

We finished 289/4243 (bronze) at private 0.942, absorbed into the public 0.950 monoculture and unable to exceed it. The value of this note is not a score but a *negative result delivered with measurements*: a falsification of five improvement directions, identification of a held-out-validation trap, a reusable measurement-gate methodology, and a platform-constraint explanation for why public foundation-model pipelines form such hard ceilings. We release all code, the cloud training/evaluation harness, and the gates openly, in the hope the map helps the next small team decide where to spend its five daily submissions.

Reproducibility & code

All model training (cloud GPU), the measurement-gate harness (`compare_to_teacher`, `eval_sed_ensemble`, the probe and rare-taxon gates), soundscape pseudo-labeling, ONNX export, and kernel-build scripts are released at: <https://github.com/Whyne-Labs/birdclef2026>. The public-baseline components are the work of the Kaggle community and are credited to their original authors: Perch v2 (Google) [1]; the EoS / Light-ProtoSSM / distilled-SED / TAX_SMOOTHING lineage (nina2025, pilkwang, tuckerarrants, hideyukizushi, chaneyma, and others). This note’s own contribution is the measurement framework and the negative-results map layered on top.

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